

Mandelkern on Modals

Cian Dorr

26 March 2018

1. The status of Wittgenstein sentences

Wittgenstein sentences:

- (1) It's raining but it might not be raining
- (2) It might be raining and it isn't raining

Observation: these sound bad in all kinds of embeddings, unlike 'It's raining and I don't know it'.

A paradigm embedding: 'Wittgenstein disjunctions'.

- (3) Either Sue might be my TA but she isn't, or Mary might be my TA but she isn't.

A tempting generalization: 'p and might q' and 'might q and p' are both interchangeable with 'p and might (p and q)', in all manner of embeddings. So in particular 'not p and might p' is interchangeable with the contradictory 'not p and might (p and not p)'.

- This isn't *quite* right: 'It might be raining and it might not be raining' seems fine, and not equivalent to the bizarre 'It might be raining and it might be that (it's not raining and it might be raining)'. But seems typically OK for simple non-modal p and q.

Some related generalizations with the same apparent status:

- (4) 'p or might q' $\Leftrightarrow$ 'p or might (not-p and q)'
- (5) 'Some F might be G' $\Leftrightarrow$ 'Some F might be F and G'
- (6) 'Something that might be F is G' $\Leftrightarrow$ 'Something that might be F and G is G'
- (7) 'Every F might be G' $\Leftrightarrow$ 'Every F might be F and G'

2. Strategies from the literature

- Dynamic semantics. Gets 'p and might q' to be equivalent to 'p and might (p and q)', but does nothing for 'might q and p'.
- "Domain" semantics. Does not secure sameness of *compositional semantic value* for 'p and might q' and , only sameness of "acceptance conditions". As a result, requires special purpose semantics for certain embedding operators (e.g. 'if', 'suppose') to predict interchangeability. Lacks a plausible semantics for 'or' that predicts interchangeability.

3. Highlights of Matt's treatment

- (i) Modals are context-sensitive. What context provides is an accessibility relation. We model this by labelling each occurrence of a modal in a sentence with a variable of

the type $s \rightarrow s \rightarrow t$ (relations among worlds); context provides an assignment function g . ‘ $\text{Might}_i p$ ’ is true at w on assignment g iff p is true on g at some world in $g(i)(w)$.

- (ii) (Typically:) When g is the assignment function provided by the context of an utterance of a sentence embedding ‘ p and $\text{might}_i q$ ’ or ‘ $\text{might}_i q$ and p ’ (where p and q are non-modal), $g(i)(w)$ is a set of p -worlds for every relevant p -world w .

NB: so long as g meets this condition, ‘ p and $\text{might}_i q$ ’ is true at a world exactly when ‘ p and $\text{might}_i (p \text{ and } q)$ ’ is (at least when p and q are nonmodal).

- (iii) The generalisation in (ii) is explained as a presupposition, implemented via undefinedness. If g is an assignment function that doesn’t fit the generalisation, $\llbracket \text{Might}_i p \rrbracket$ will just be undefined on g at every world.

- Note: this is a particularly straightforward use of the gap-based theory of presupposition. We need never be in a position of saying that a sentence is uttered in a context relative to which it’s gappy in some worlds and true or false in others, so we don’t have to wrestle with the question what speech act one would perform by uttering such a sentence. Presupposition merely takes certain interpretative options off the table.

- (iv) The presuppositions in (iii) are implemented by assigning sentences truth values only relative to a triple g, κ, w , where g is an assignment function, κ is a set of worlds, and w is a world. We have

$$\begin{aligned} \llbracket \text{Might}_i p \rrbracket(g, \kappa, w) \text{ is } & \text{a. defined only when } \forall w' \in \kappa : g(i)(w') \subseteq \kappa \\ & \text{b. if defined, } = 1 \text{ iff } \exists w' \in g(i)(w) : \llbracket p \rrbracket(g, \kappa, w') = 1 \end{aligned}$$

An asserted sentence should be defined when κ is the *global context* (set of worlds compatible with what everyone’s taking for granted).

- At least in this fragment treated, κ only matters for definedness: a sentence that’s true relative to g, κ, w can’t be false relative to g, κ', w (though it may be undefined).
- Given a Weak Kleene treatment of connectives (‘undefinedness is infectious — see note 19), a sentence that’s undefined relative to g, κ, w is also undefined relative to g, κ, w' for every w' .
- So, a different way to achieve the same effect would be to assign truth values only relative to an assignment and a world in the standard way, and give a separate recursive definition of ‘assignment g is admissible for sentence p relative to context κ ’.
- This reformulation reveals a more demanding sense in which the logic is classical: classically valid arguments preserve truth on every assignment (and also on every assignment that’s admissible for all sentences involved in some context). What Matt says is that classical rules (like conjunction introduction) are *Strawson-*

valid: they preserve truth whenever all presuppositions are satisfied. But Strawson-valid arguments don't always seem so great: 'He doesn't regrets anything he did; so he doesn't regret drinking a can of paint', or 'Every animal in my house is purring; so the elephant in my house is purring'.

- (v) The extra parameter κ tracks the notion of the *symmetric local context* of a constituent formula, as determined by Schlenker's algorithm — intuitively, whatever the strongest proposition is that could be conjoined to that constituent formula without changing the set of worlds in the context where the sentence is true. For example:

$$\begin{aligned}\llbracket p \text{ and } q \rrbracket(g, \kappa) &= \llbracket p \rrbracket(g, \kappa \cap \llbracket q \rrbracket(g, \kappa)) \cap \llbracket q \rrbracket(g, \kappa \cap \llbracket p \rrbracket(g, \kappa)) \\ \llbracket p \text{ or } q \rrbracket(g, \kappa) &= \llbracket p \rrbracket(g, \kappa \setminus \llbracket q \rrbracket(g, \kappa)) \cap \llbracket q \rrbracket(g, \kappa \setminus \llbracket p \rrbracket(g, \kappa))\end{aligned}$$

Or, in my proposed reformulation:

g is admissible for ' p and q ' relative to κ iff g is admissible for p relative to $\kappa \cap \llbracket q \rrbracket^g$
and g is admissible for q relative to $\kappa \cap \llbracket p \rrbracket^g$
 g is admissible for ' p or q ' relative to κ iff g is admissible for p relative to $\kappa \setminus \llbracket q \rrbracket^g$
and g is admissible for q relative to $\kappa \setminus \llbracket p \rrbracket^g$

In the case of quantified sentences, local contexts are assignment-relative; for attitude reports they are world-relative.

g is admissible for 'Some p is q ' relative to κ iff for some x , g is admissible for p relative to $\kappa \cap \llbracket q \rrbracket^g(x)$ and g is admissible for q relative to $\kappa \cap \llbracket p \rrbracket^g(x)$.
 g is admissible for 'A believes p ' relative to κ iff for every $w \in \kappa$, g is admissible for p relative to the set of all worlds compatible with what A believes in w .

4. Strong versus weak versions

Matt considers two versions of the undefinedness condition for 'Might'.

Strong Boundedness: $\llbracket \text{Might}_i p \rrbracket(g, \kappa, w)$ is undefined only if $\forall w' : g(i)(w') \subseteq \kappa$

Weak Boundedness: $\llbracket \text{Might}_i p \rrbracket(g, \kappa, w)$ is undefined only if $\forall w' \in \kappa : g(i)(w') \subseteq \kappa$

My previous argument against Strong Boundedness turned on

Reflexivity: $\llbracket \text{Might}_i p \rrbracket(g, \kappa, w)$ is undefined only if $\forall w' : w' \in g(i)(w')$.

Combining Reflexivity with Strong Boundedness means that $\llbracket \text{Might}_i p \rrbracket(g, \kappa, w)$ is defined only when κ is the set of all worlds, absurdly.

Why would you want Reflexivity? One argument: it explains why 'Must p but not p ' and ' p but not might p ' are bad. But this could be explained by

Weak Reflexivity: $\llbracket \text{Might}_i p \rrbracket(g, \kappa, w)$ is undefined only if $\forall w' \in \kappa : w' \in g(i)(w')$.

which is consistent with Strong Boundedness. So we need to look at embeddings. Previously I offered

(8) Either John is here and must_i be upstairs, or Susie is here and must_j be upstairs

This seems like it should entail

(9) Either John is here and is upstairs, or Susie is here and is upstairs

But with only Weak Reflexivity, (8) can be true at w_3 in the following 3-world context:

w_1 : only John is here, he's upstairs

w_2 : only Susie is here, she's upstairs

w_3 : both John and Susie are here, they're downstairs

$g(i)(w) = \{w_1\}$ for every w

$g(j)(w) = \{w_2\}$ for every w .

5. Two more arguments against Strong Boundedness

An argument from ellipsis. Consider:

(10) Either it's a pool party and John is unlikely_i to be having fun, or it's a fancy dress party and Mary is unlikely_j to be having fun.

Given Strong Boundedness, (10) can be true relative to g, κ, w only if $g(i)(w)$ is a set of worlds where it's a pool party (in few of which John is having fun) and $g(j)(w)$ to be a set of worlds where it's a fancy dress party (in few of which Mary is having fun). So it's got to be that $i \neq j$.

But note that we can seamlessly ellipse the last bit of (10):

(11) Either it's a pool party and John is unlikely to be having fun, or it's a fancy dress party and Mary is.

Given standard lore about ellipsis, we're not allowed to 'reconstruct' using a different variable.

An argument from quantification: Consider

(12) Someone who might_i be a gracious winner is the winner

Given Strong Boundedness, g is admissible for (12) relative to κ only if $\exists x : g(i)(w')$ is a set of worlds in κ in which x is the winner.

So: if there's no x who we know to be the winner, there will be no g that's admissible for (12) relative to the global context such that we know the proposition expressed by (12) relative to g !

- Suppose for example that there are two possible winners, Janet and Mary. Then if g is an assignment for which $g(i)$ maps every world to a world where Janet is a winner,

(12) is false in a world where Mary is the winner, while if $g(i)$ maps every world to a world where Mary is a winner, (12) is false in a world where Janet is the winner. On neither interpretation does (12) express something we know to be true.

By contrast, given Weak Boundedness, (12) is fine so long as for each relevant w , $g(i)(w)$ is a set of worlds that agree with w as regards who the winner is.

6. What's the advantage of Strong Boundedness?

Matt mentions the fact that it makes available a “restrictor” account of conditionals, where ‘if’ clauses merely reduce the local context (thereby forcing restricted interpretations for modals):

$$\llbracket \text{if } p, q \rrbracket(g, \kappa) = \llbracket q \rrbracket(g, \kappa \cap \llbracket p \rrbracket(g, \kappa))$$

I'm dubious about this program.

- How can be extended to deal with counterfactual conditionals? Whatever is going on with ‘would’ it's clearly not forced to be restricted to its local context.
- How is it going to explain the familiar interaction of ‘if’ with deontic modals, as in ‘If you're going to kill her you should kill her gently’ or ‘If you're going to Harlem you should take the A train’? There can't be a requirement for deontic modals to be restricted to their local contexts, since ‘He's not going to Harlem but he should be’ is fine.

7. Unconditionals

I claim that on its natural reading, (13) is equivalent to (14):

(13) Whether or not it's raining, Jim might be carrying an umbrella

(14) If it's raining Jim might be carrying an umbrella, and if it's not raining Jim might be carrying an umbrella.

Both in some sense imply

(15) It might be that it's raining and Jim's carrying an umbrella, and it might be that it's not raining and Jim's carrying an umbrella.

Similarly for (16) and (17):

(16) Whoever comes to the party, Jim will probably have a good time

(17) If only Matt comes to the party Jim will probably have a good time, and if Bar and Vera come to the party Jim will probably have a good time, and...

It's not obvious how Matt's system can predict this. Applying the Schlenker algorithm suggests that the local context of q in ‘Whether or not p , q ’ is just the the global context, in which case there'll be nothing to rule out interpreting the ‘might’ using an accessibility relation where the worlds in the global context all see one another.

Possible proposal: $\llbracket \text{Whether or not } p, q \rrbracket^{g, \kappa, w} = \llbracket \text{If } p, q \text{ and if not } p, q \rrbracket^{g, \kappa, w}$. So q ends up being evaluated twice, first relative to a local context κ' that entails p , and then relative to a local context κ'' that entails $\text{not-}p$.

- Note that this will only work given Weak Boundedness. Strong Boundedness makes, 'If p , $\text{might}_i q$ and if not p , $\text{might}_i q'$ is inconsistent (although 'If p , $\text{might}_i q$ and if not p , $\text{might}_i q'$ is consistent for $i \neq j$).

Problem: if that's how unconditionals work, why is there such a difference between (18) (fine) and (19) (pretty weird)?

(18) Whether or not it's raining, it might be raining.

(19) If it's raining it might be raining, and if it's not raining it might be raining.

8. Adding factives to Wittgenstein sentences

Sentences like (20) seem as odd as normal Wittgenstein sentences, and remain odd under many embeddings, as witness (21):

(20) It's not raining and A knows it might be.

(21) Either it's not raining and she knows it might be, or it's raining and she knows it might not be.

Neither version of Matt's semantics helps explain this. The local context for q in 'p and A knows q ' does not entail p , since 'p and A knows q ' is not equivalent to 'p and A knows (p and q)'.

Suggestion: what's needed is something more flexible than the machinery of local contexts, that makes it natural for a modal in one conjunct to be 'constrained' by the other even when it's under some further operator like 'knows'. But no matter what the differences between factive and non-factive attitudes looks mysterious.

9. Tenses on epistemic modals

In English, modal auxiliaries don't carry tense marking in the usual way. 'She might have been self-employed' is ambiguous between 'It is possible that she was self-employed' and 'It was possible that she was self-employed', and perhaps also 'It is possible for her to be self-employed'. But other languages don't have such an ambiguity, and we can avoid it by focusing on 'be possible'. (Note that 'It is possible that it is raining' only has an epistemic reading!)

Clearly, 'be possible' is time-relative in whatever way 'be tired' is. How to add this to a basic relational framework? A natural strategy: 'possible' combines with (a) a variable over functions from *instants* and worlds to sets of worlds, and (b) an element (tense morpheme?) denoting an instant:

(If defined:) $\llbracket \text{It is}_t \text{ possible}_i \text{ that } p \rrbracket^{g, w} = 1$ iff $\exists w' \in g(i)(g(t))(w) : \llbracket p \rrbracket^{g, w} = 1$

Option 1: defined for g, κ, w only if $\forall t : \forall w' \in \kappa : g(i)(t)(w') \subseteq \kappa$

Option 2: defined for g, κ, w only if $\forall w' \in \kappa : g(i)(g(t))(w') \subseteq \kappa$

Option 3: defined for g, κ, w only if $\forall w' \in \kappa : g(i)(\text{present time})(w') \subseteq \kappa$

- More suggestions?

The OKness of (22) suggests that options 1 and 2 are too demanding for past tense examples:

- (22) Yesterday morning it was raining although it was possible that it was dry. (So we ended up leaving our umbrellas at home and getting wet.)

Maybe this is some special past-tense thing? But (23) and (24) are also not so bad...

- (23) Every morning when it's raining and it's possible that it's dry, we end up leaving our umbrellas at home and getting drenched.
- (24) Tomorrow morning, if it's dry and it might be raining, we'll bring umbrellas but have no use for them.

Is it just Option 3? But consider (25):

- (25) Whenever it's raining and it's possible that it's dry, the weather monitors are broken.

This doesn't imply 'the weather monitors are broken'!

I find it hard to see how any theory structurally like this can be got to work once we include tense in the semantics, as we must. Reason to hope for a pragmatic story!